Expectation Propagation for Approximate Bayesian Inference

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M2 MVA

March 12, 2026

The Bayesian problem

$D = \{y_1, \ldots, y_n\}$, we know $p(x)$ and $p(y_i \mid x)$, and we search for $p(D)$ and $p(x \mid D)$.

Assumed-Density Filtering (ADF):

- We choose an approximation family of functions $\mathcal{F}$ (e.g. Gaussians)
- We iteratively construct $q(x)$ in $\mathcal{F}$ approximating $p(x, D)$:

1. We start with $q_0(x) = p(x)$
2. For $i = 1, \ldots, n$:

We multiply $q_{i-1}(x)$ by $p(y_i|x)$, we normalize and we approximate in $\mathcal{F}$ to obtain $q_i(x)$.

At each step we have produced a normalizing constant Z_i and the product of those Z_i gives an approximation of $p(D)$.

Expectation Propagation and the Clutter Problem

The big issue: it depends on the order of the y_i!

The solution: Expectation Propagation (EP)

- Instead of approximating $q(x) = p(x, D)$, we approximate $p(y_1|x), \ldots, p(y_n|x)$
- We will use ADF steps inside EP

The clutter problem

- We choose $p(x)$ to be a Gaussian, $\mathcal{F}$ is the set of n-dimensional Gaussians

$$p(x) = \mathcal{N}(x; m_0, \Sigma_0)$$

- $p(y_i|x)$ will be a mixture of two Gaussians

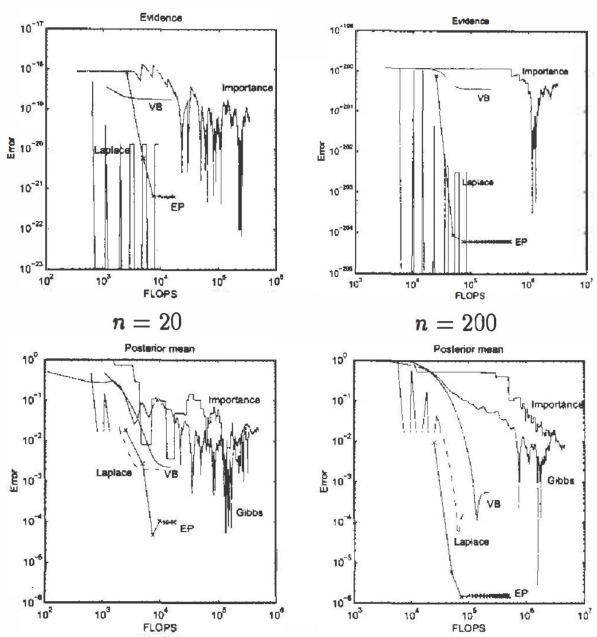
$$p(y|x) = (1 - w)\mathcal{N}(y; x, I) + w\mathcal{N}(y; 0, \Sigma), \quad w \in (0, 1)$$

In this case all is computable, and we did the computations in a **more general case** than the one in the article.

Method Comparison (Evidence and Posterior Mean)

Methods compared:

- Expectation Propagation
- Laplace's method
- Variational Bayes
- Importance Sampling
- Gibbs Sampling



$n = 20$ $n = 200$

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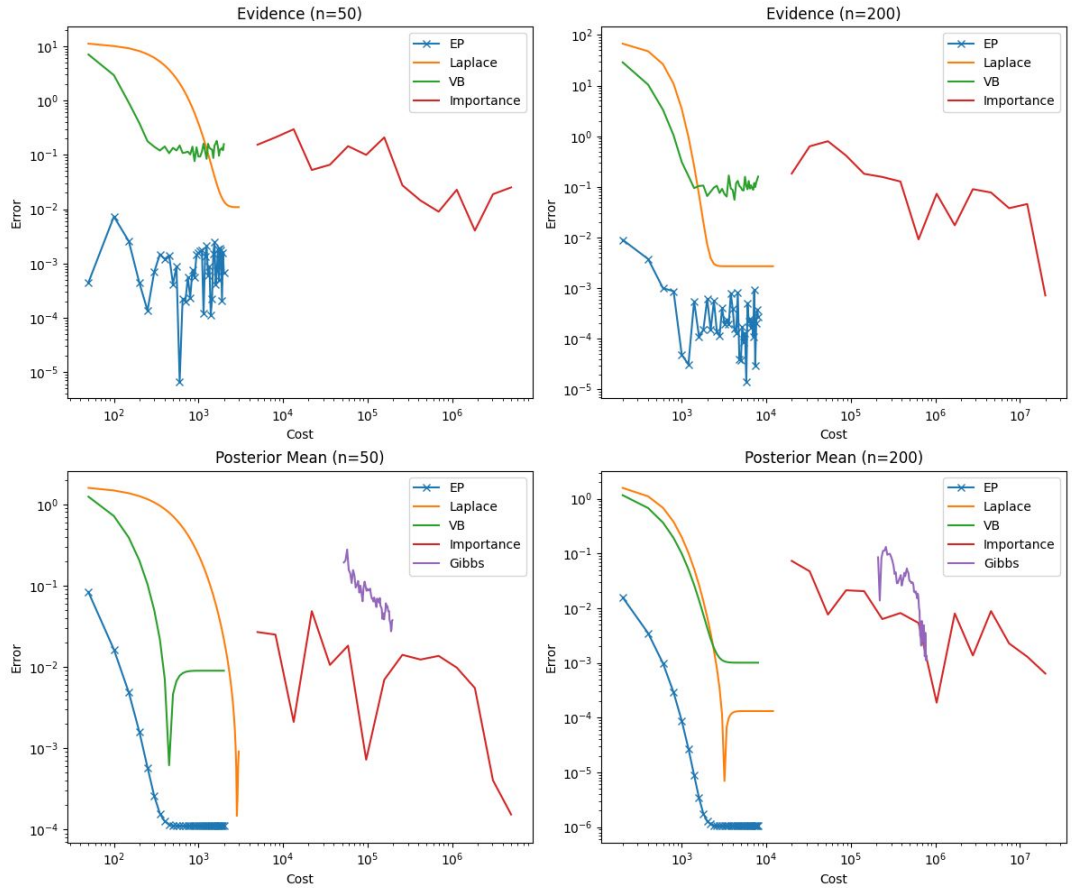
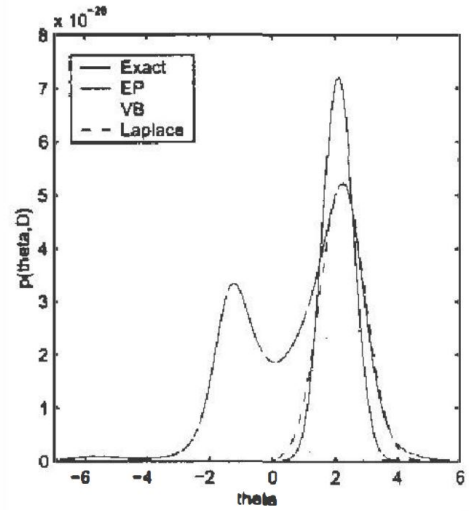
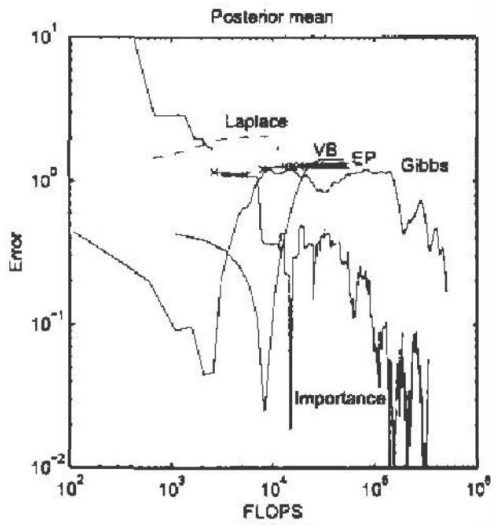
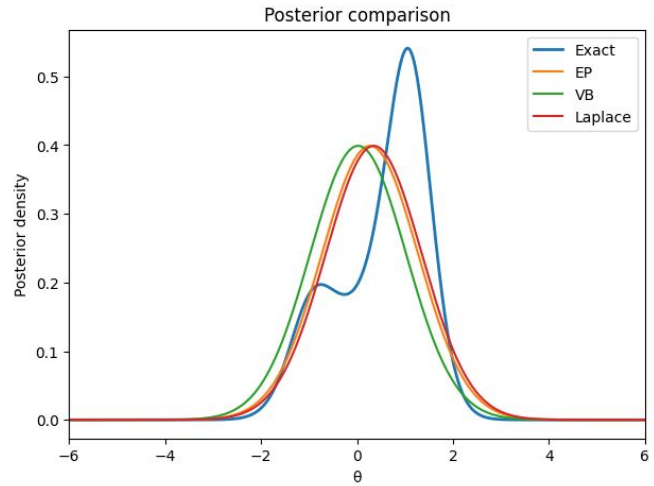
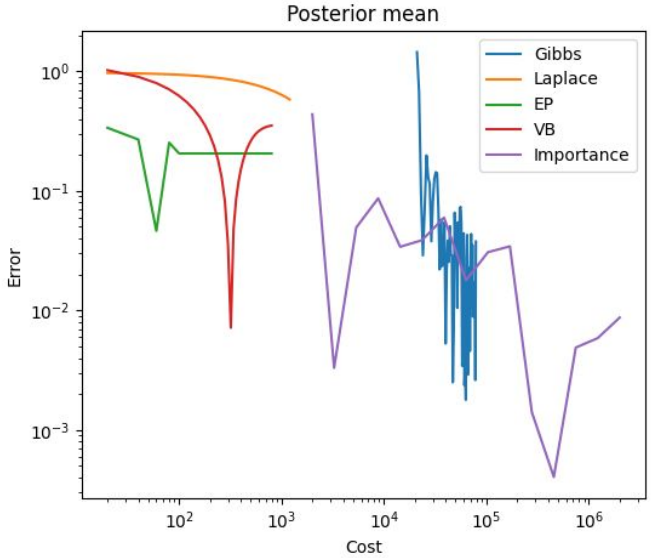


Potential Drawback



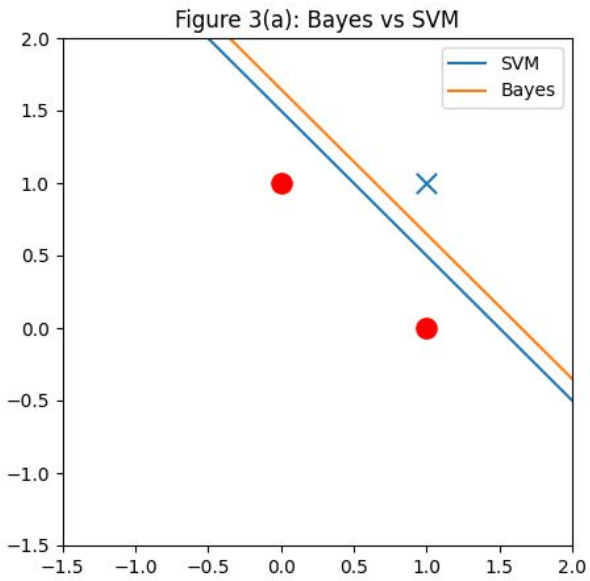


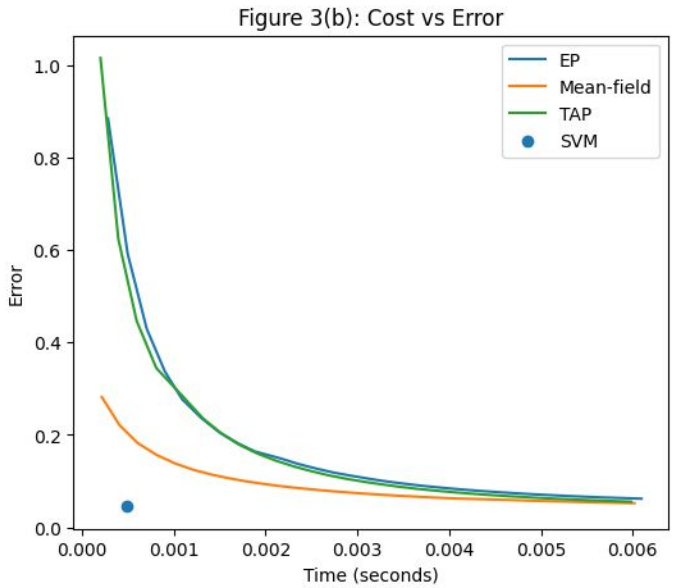




Bayes Point Machine

- A Bayesian linear classifier using a weight vector to separate two classes.
- The Bayesian solution averages over classifiers weighted by their posterior probability.
- In practice, BPM uses the average weight vector (the Bayes point) for prediction.
- Expectation Propagation (EP) approximates the posterior over the weights with a Gaussian.





Reference

Expectation Propagation for Approximate Bayesian Inference
Thomas P. Minka

https://arxiv.org/pdf/1301.2294

Gitlab
https://gitlab.com/vibhud04/expectation-propagation-for-approximate-bayesian-inference

Bayes Point Machine

$$p(w) = \mathcal{N}(w; 0, D)$$

$$p(y_i | z_i, w) = \sigma\left(\frac{y_i w^\top z_i}{\varepsilon}\right)$$

where

$$\sigma(z) = \int_{-\infty}^{z} \mathcal{N}(t; 0, 1)\, dt$$